



EDUCATION

Bachelor of Computer Engineering, Major in Game Technology

Electronic Engineering Polytechnic Institute of Surabaya (2020 - 2024)
GPA 3.67 (Cumlaude)

TECHNICAL SKILLS

- Python (NumPy, scikit-learn, PyTorch)
- Reinforcement Learning (Q-Learning, FSM)
- Fuzzy Logic & Recommender Systems
- Data Scraping, Analysis, Visualization (Selenium, Gephi, Tableau)
- C++ | C# (engine-side AI integration)
- Applied research & evaluation studies

CONTACT

 Kenan Firmansyah

 Kenanfir

 nandev.dev

 +6285175447540

 realkenanfir@gmail.com

Kalijudan, Mulyorejo, Surabaya,
Jawa Timur 60114

Holland Village Jakarta, Cemp.
Putih, Jakarta Pusat, DKI Jakarta
10510

KENAN FIRMANSYAH

AI / ML ENGINEER

Applied AI/ML engineer with five peer-reviewed publications on adaptive game AI, dynamic difficulty, and context-aware recommender systems. Industry experience as a data scientist plus production AI work in shipped interactive titles.

SELECTED PUBLICATIONS

Adaptive Radio System Using MetaSounds for Serious VR Game
IEEE Access (Vol. 13), Nov 2, 2025 | DOI: 10.1109/ACCESS.2025.3627544

- FSM-driven adaptive audio across affective states; n=30 VR user study, rated more immersive than non-adaptive baseline.

Dynamic Level of Difficulties Using Q-Learning and Fuzzy Logic
IEEE Access (Vol. 12), Sep 12, 2024 | DOI: 10.1109/ACCESS.2024.3457801

- Hybrid Q-learning + fuzzy DDA system; **+35% session length, +28% player effectiveness.**

Context-Aware Recommender System for In-Game Radio
INASS IJIES, Sep 2, 2024 | DOI: 10.22266/ijies2024.1031.26

- Deep-learning hybrid filtering on contextual signals; **F1 up to 0.71**, outperforming baselines.

EXPERIENCE

Game Developer (AI/Gameplay)
Agate | 2026 Feb - 2026 May

- Production game-AI work at one of Indonesia's largest studios on an unannounced title.

Data Scientist (Intern)
PT. eBdesk | 2023, 6 months

- Built scraping, analysis, and visualization pipelines in Python (Selenium, Gephi, Tableau).

ADDITIONAL RESEARCH & APPLIED AI WORK

1 APPLIED AI IN SHIPPED & RESEARCH PROJECTS

Xanthous — AI in VR Horror (Thesis)

EEPIS | 2023 Feb - 2024 July | Unreal Engine

- Designed and implemented enemy AI systems and gameplay mechanics for a serious VR horror game; underlying research published.

Until Dawn Do Us Part — Enemy AI & System Programmer

iPlay Apple Developer Academy Game Jam 2025 | 1st Place

- Wave-based enemy AI and game systems for a survival twin-stick shooter, shipped in 2 days.

AI Programmer roles in Game Jams

Brackeys 2024.2 (Unreal AI) | Garena GameJam (Unreal AI Perception, stealth)

- Specialized AI/perception work across jam-scale prototypes.

2 ADDITIONAL PUBLICATIONS

Toward Adaptive VR Design: Physiological and Cognitive Predictors of Cybersickness in Serious Games

IJHCI, Taylor & Francis | 2026 | DOI: 10.1080/10447318.2026.2684064

Multivariate cybersickness model in a UE5 serious game (n=105). $R^2 = 0.358$; duration, cognitive load, and GSR as top predictors for adaptive comfort systems.

Real Time Procedural Audio for Immersive Gameplay with MetaSounds

TCCE 2025, Trends in Computational and Cognitive Engineering

Dynamic Day and Night Cycle Impact in a Serious VR Game

ICVR 2024, International Conference on Virtual Reality | +25% immersion improvement reported.

3 OTHER ENGINEERING EXPERIENCE

Apple Developer Academy @ BINUS (2025 Feb - 2025 Dec) — Interdisciplinary product development; UI/UX, prototyping, iterative testing.

Lead Developer, Starpixel Studios (2024 Oct - 2025 April) — Multiplayer horror game; Blueprint + C++ for performance-critical systems.

JoyBait Studio (Saturated) (2025 Oct - 2025 Dec) — Project manager and programmer on a first-person horror game shipped to Mac App Store.

Home Server (Personal, 2023) — Proxmox-based multi-purpose server (NAS, database, compute).

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